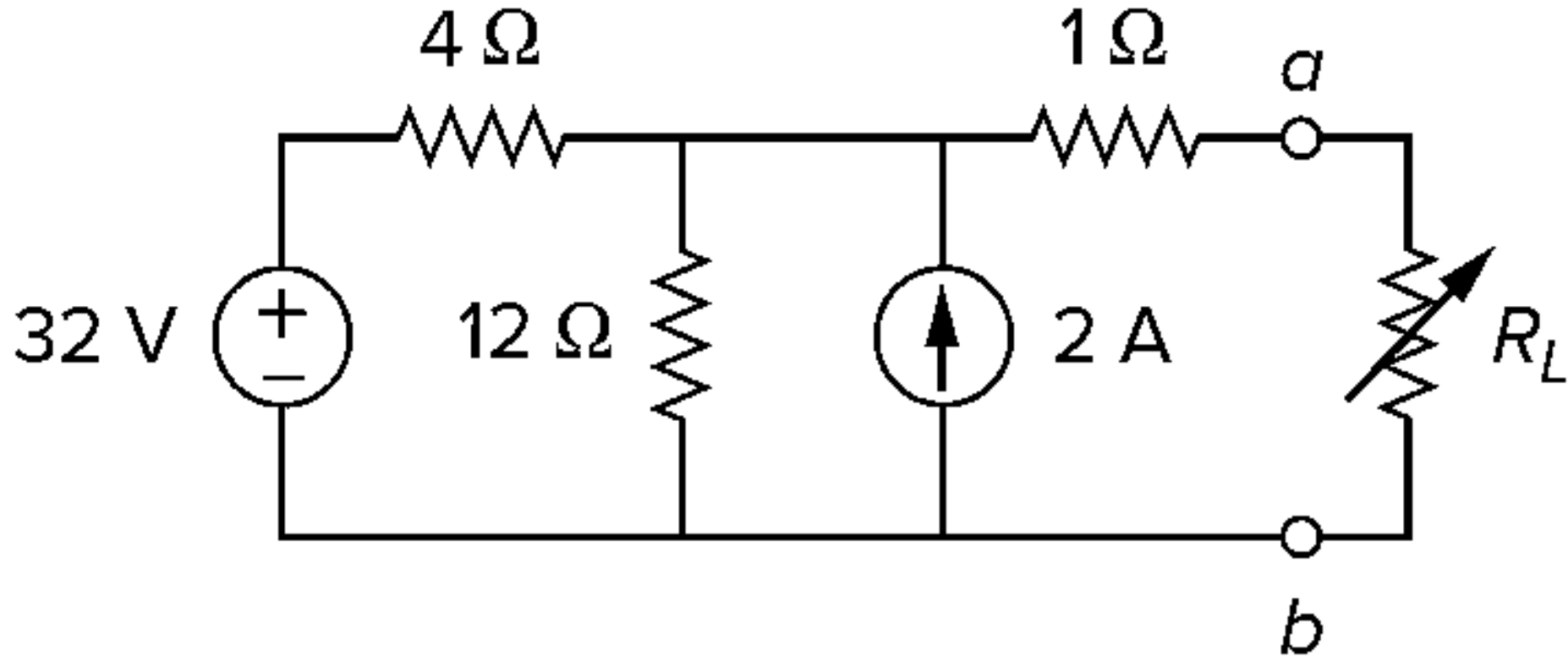


NUMERICAL

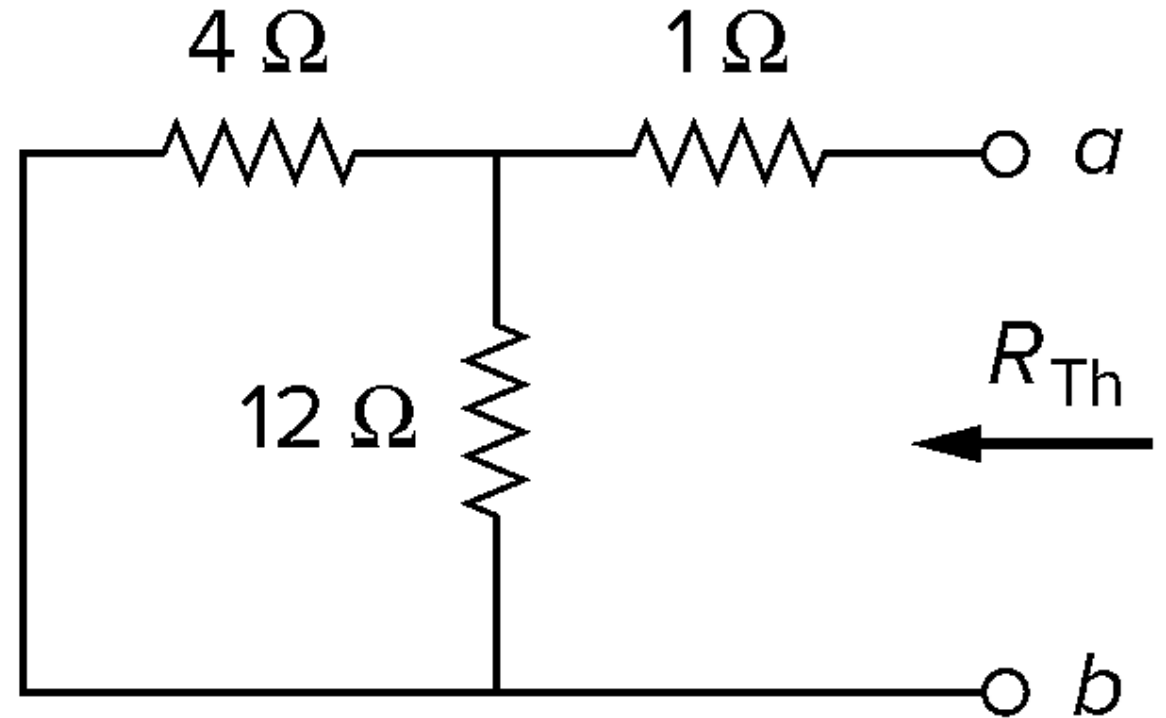
Q1. Find the Thevenin equivalent of the circuit shown below across terminals a - b . Then find the current through $R_L = 6\Omega$ and 36Ω respectively.



NUMERICAL

Soln:

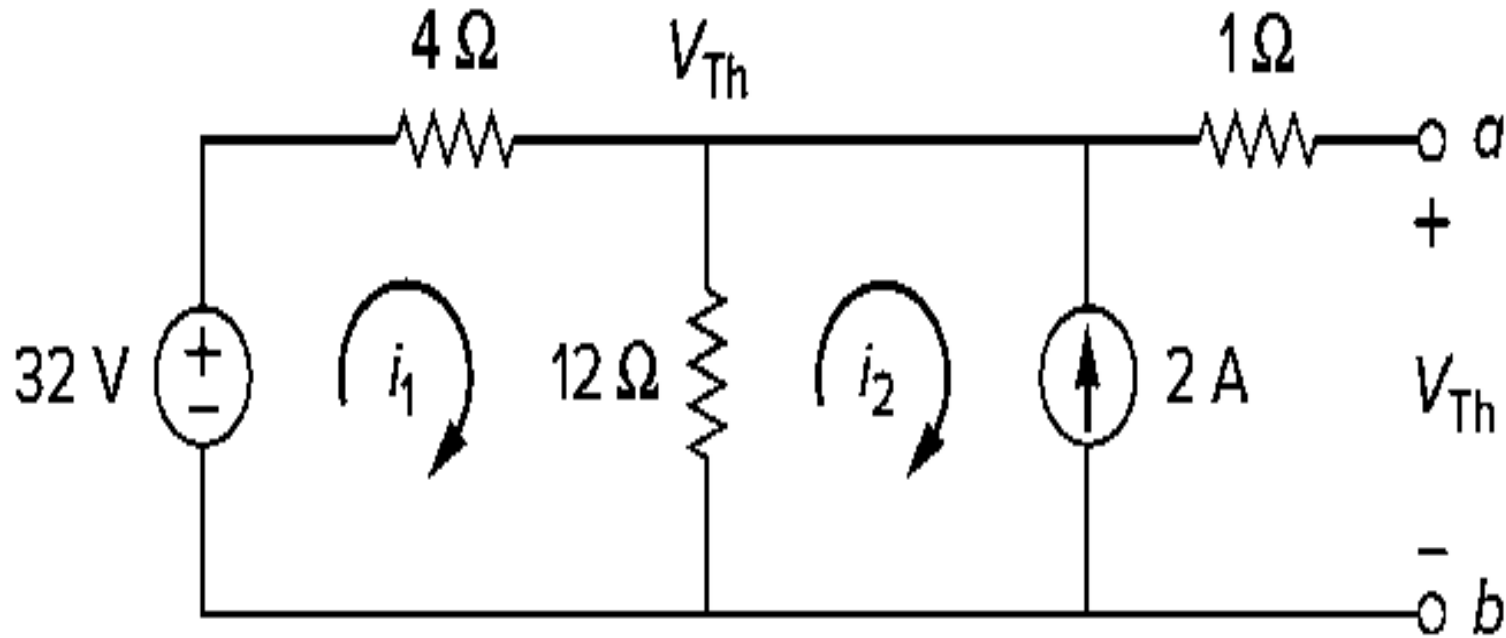
Step 1. Find R_{Th} by turning off the 32V voltage source (replacing it with a short circuit) and the 2A current source (replacing it with open circuit).



$$\boxed{R_{Th}} = 4 \parallel 12 + 1 = \frac{4 \times 12}{16} + 1 = \boxed{4\Omega}$$

NUMERICAL

Step 2. Make a - b open circuit. Find V_{Th} by applying mesh/node analysis.



$$-32 + 4i_1 + 12(i_1 - i_2) = 0, i_2 = -2A \Rightarrow i_1 = 0.5A$$

$$\boxed{V_{Th}} = 12(i_1 - i_2) = 12(0.5 + 2) = \boxed{30V}$$

NUMERICAL

Step 3. Finding current through R_L

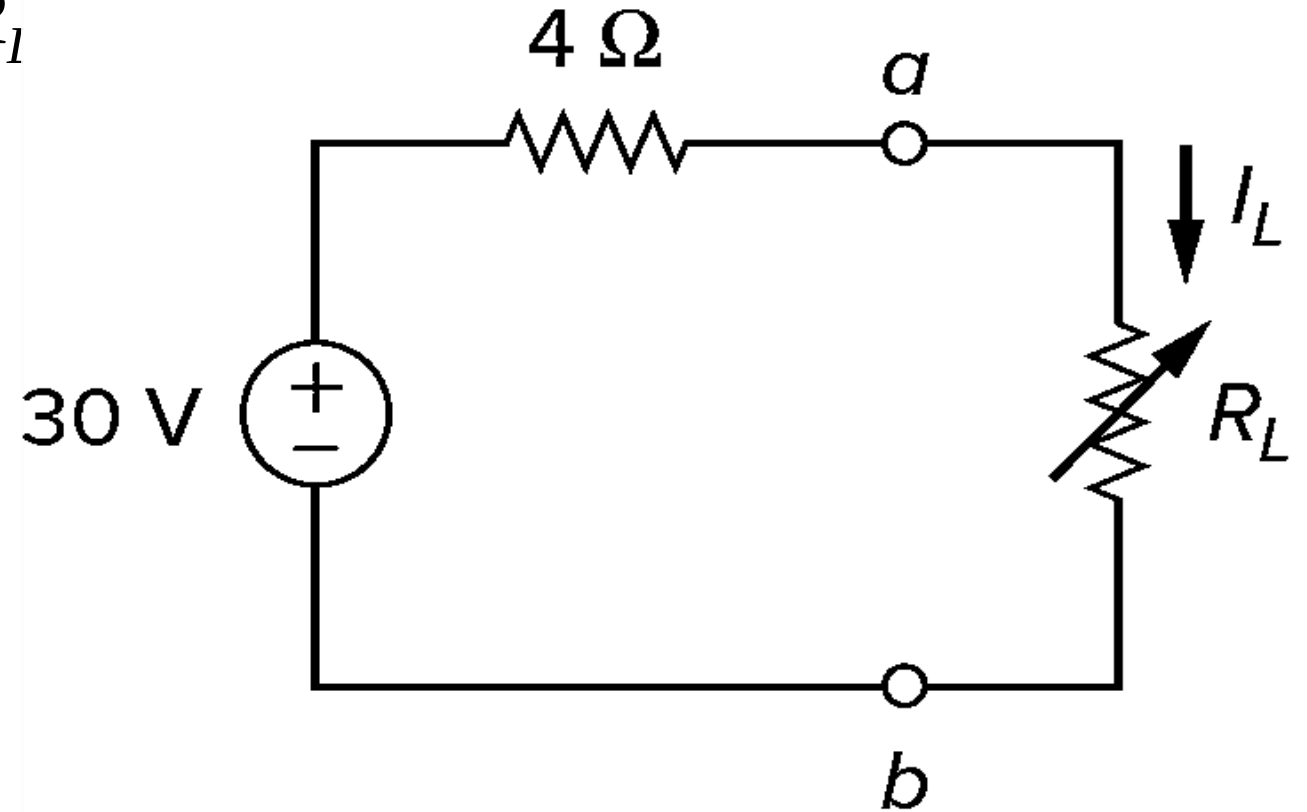
$$I_L = \frac{V_{Th}}{R_{Th} + R_L}$$

a) When $R_L = 6\Omega$

$$I_L = \frac{30}{4 + R_L} = \frac{30}{10} = 3A$$

a) When $R_L = 36\Omega$

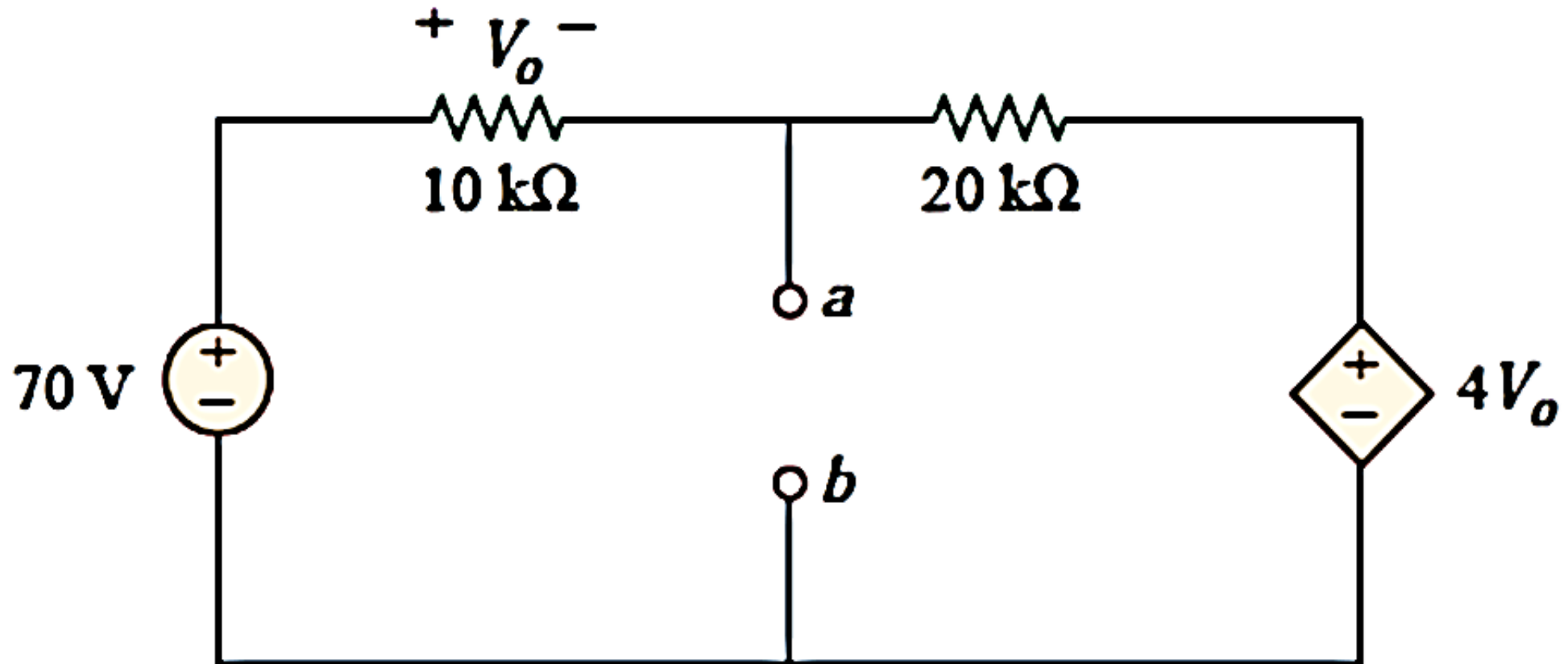
$$I_L = \frac{30}{4 + R_L} = \frac{30}{40} = 0.75A$$



Thevenin equivalent circuit

NUMERICAL

Q3. Find the Thevenin equivalent of the given circuit at terminals a - b .



NUMERICAL

Soln:

Step 1. Finding V_{Th} .

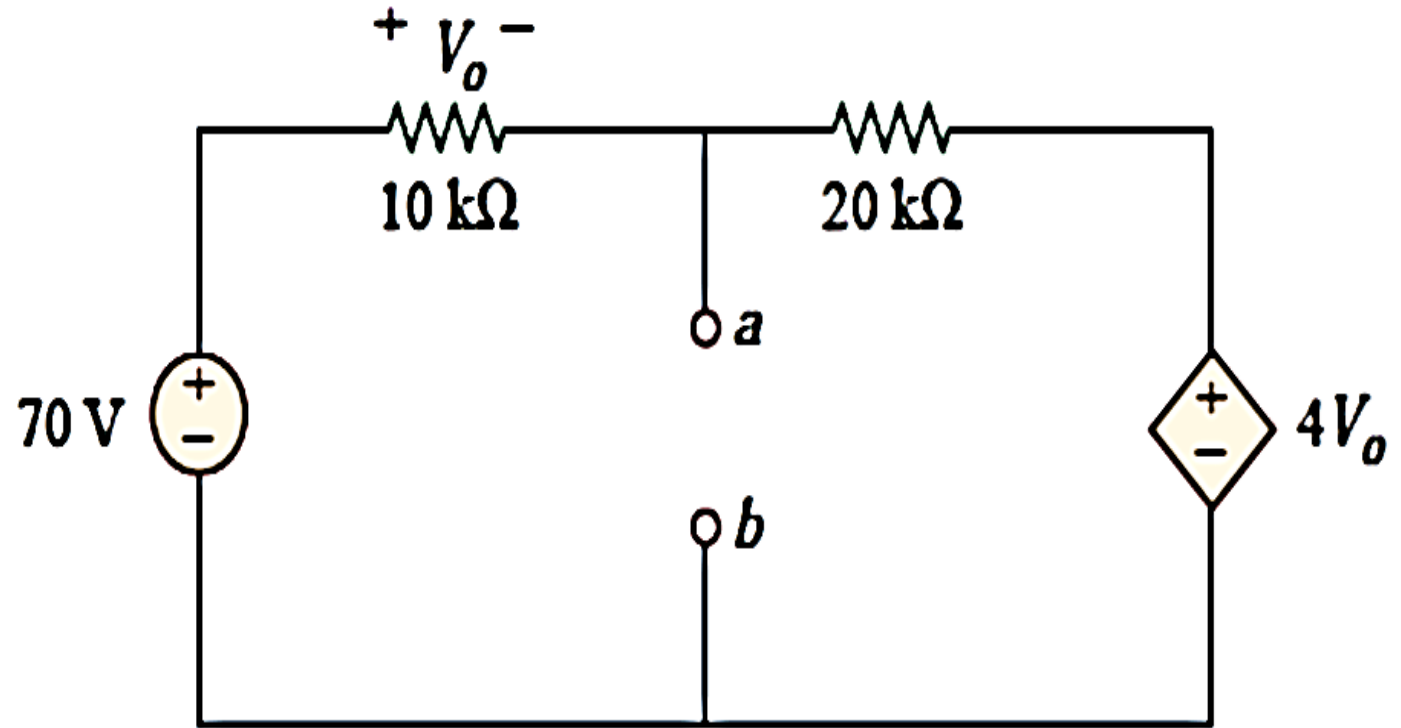
Applying KVL to the loop,

$$-70 + (10 + 20)kI + 4V_o = 0$$

But $V_o = 10k I$,

$$70 = 70kI \Rightarrow I = 1mA$$

$$-70 + 10kI + V_{Th} = 0 \Rightarrow V_{Th} = 60V$$



NUMERICAL

Step 2: To find R_{Th} , remove independent voltage source 70V and apply a 1-V independent source to excite the circuit at the terminals $a-b$

We have

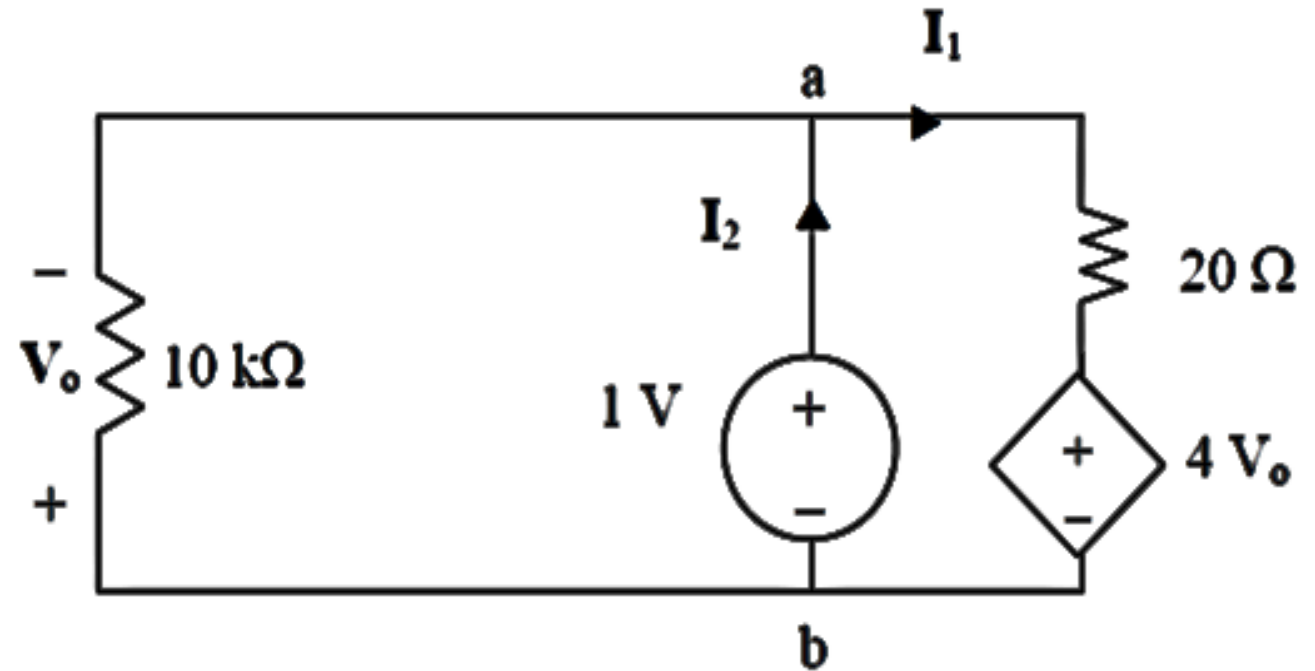
$$V_0 = -1V$$

$$-1 + 20kI_1 + 4V_0 = 0$$

$$\Rightarrow I_1 = 0.25mA$$

And

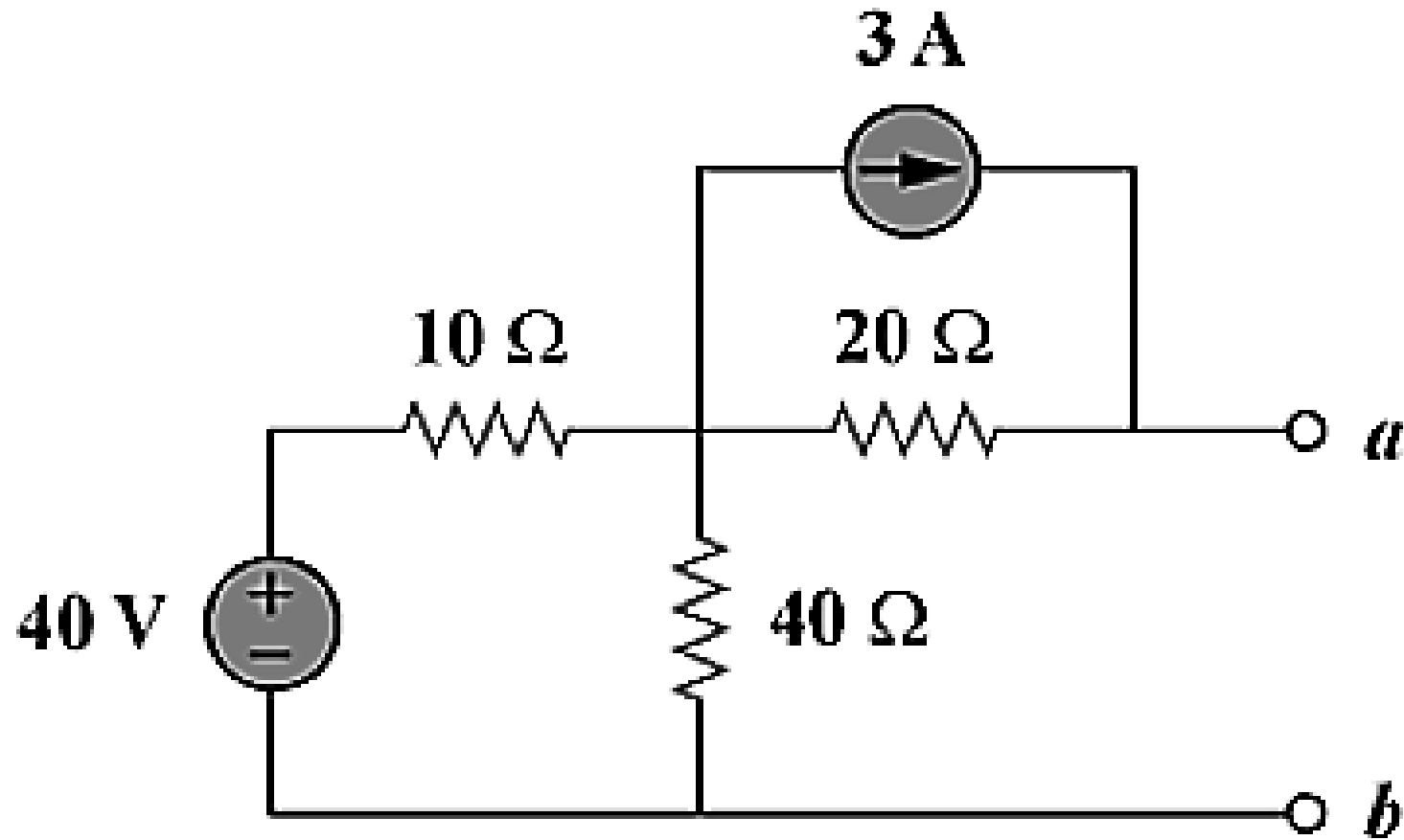
$$I_2 = I_1 + \frac{1}{10 \times 10^3} = 0.35mA$$



$$\boxed{R_{Th}} = \frac{1V}{I_2} = \frac{1}{0.35} k\Omega = \boxed{2.857k\Omega}$$

NUMERICAL

Q4. Determine the Thevenin equivalent circuit between terminals a - b .



NUMERICAL

Soln:

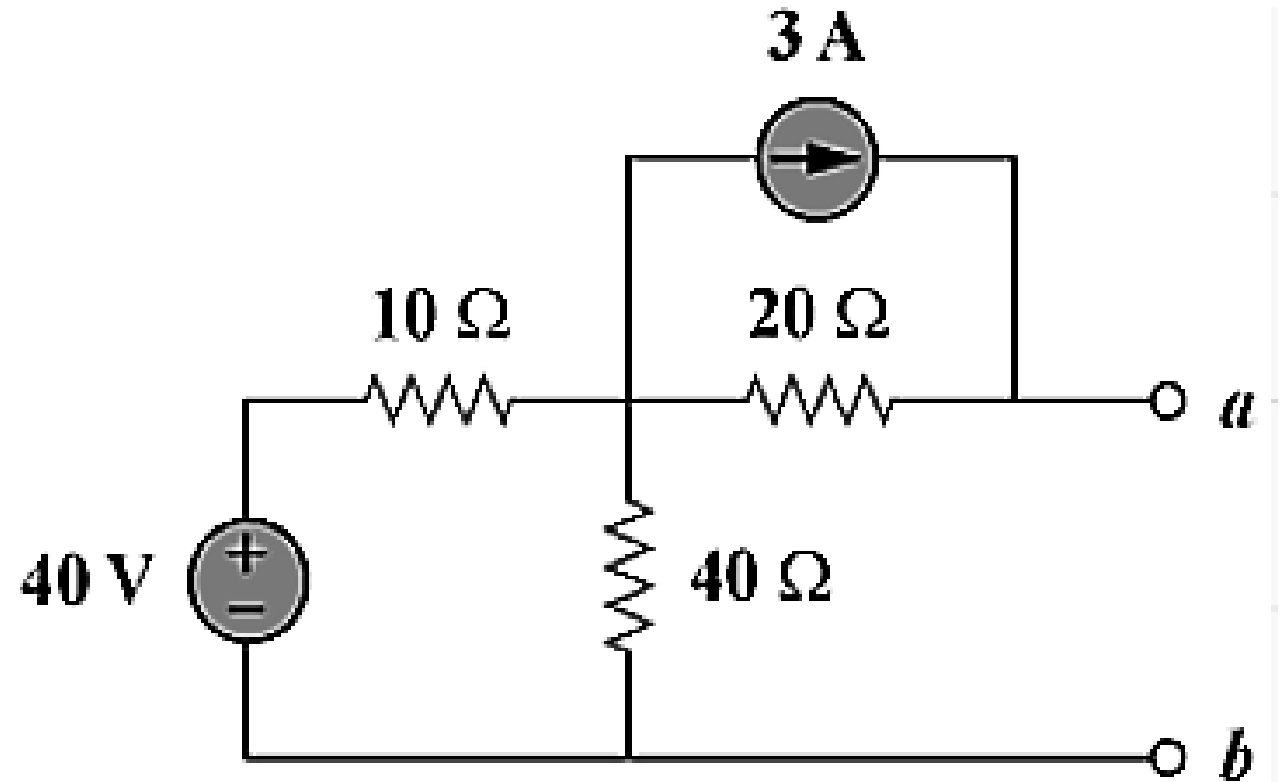
Step 1: Finding R_{Th} .

Remove independent sources
i.e replace voltage source by
short circuit and current
source by open circuit.

Then we have,

$$R_{Th} = 20 + 10 \parallel 40 = 20 + 400 / 50$$

$$\Rightarrow R_{Th} = 28\Omega$$



NUMERICAL

Step 2: Finding V_{Th} .

KCL at node 1:

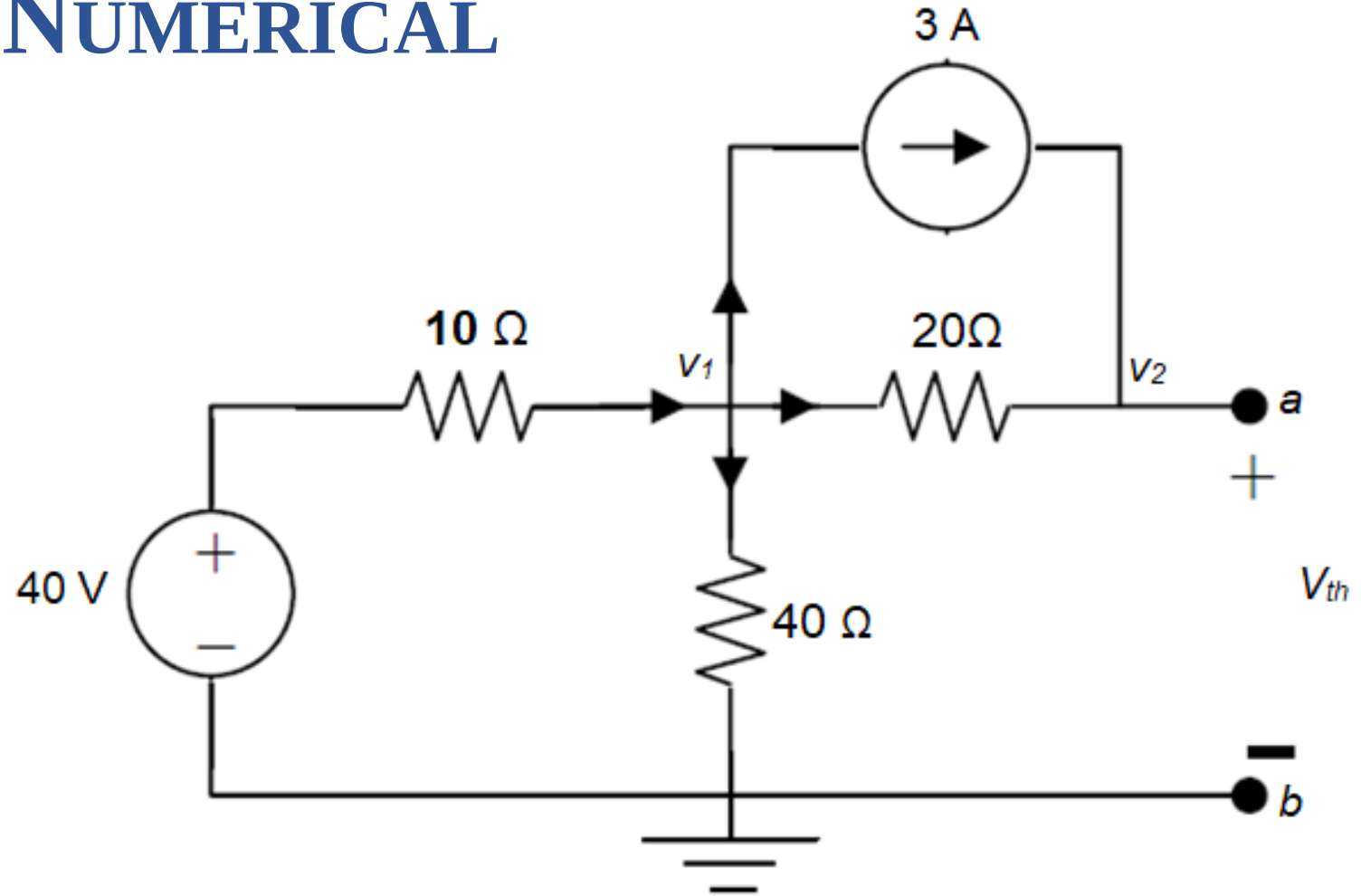
$$\frac{40 - v_1}{10} = 3 + \frac{v_1 - v_2}{20} + \frac{v_1}{40}$$
$$\Rightarrow 40 = 7v_1 - 2v_2 \quad (1)$$

KCL at node 2:

$$3 + \frac{v_1 - v_2}{20} = 0$$
$$\Rightarrow v_1 = v_2 - 60 \quad (2)$$

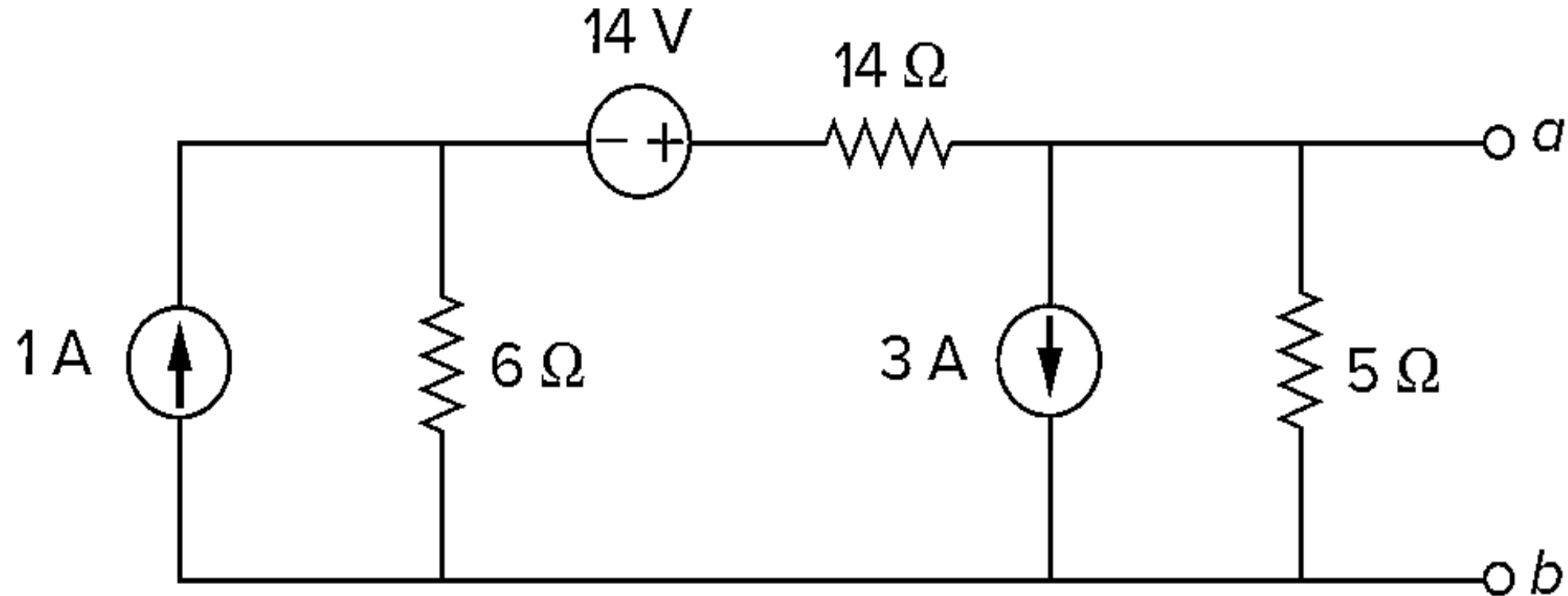
Solving (1) and (2), $v_1 = 32\text{V}$
 $v_2 = 92\text{V}$

$$\Rightarrow V_{Th} = v_2 = 92\text{V}$$



NUMERICAL

Q5. Find the Thevenin and Norton equivalents at terminals a - b of the circuit shown below.



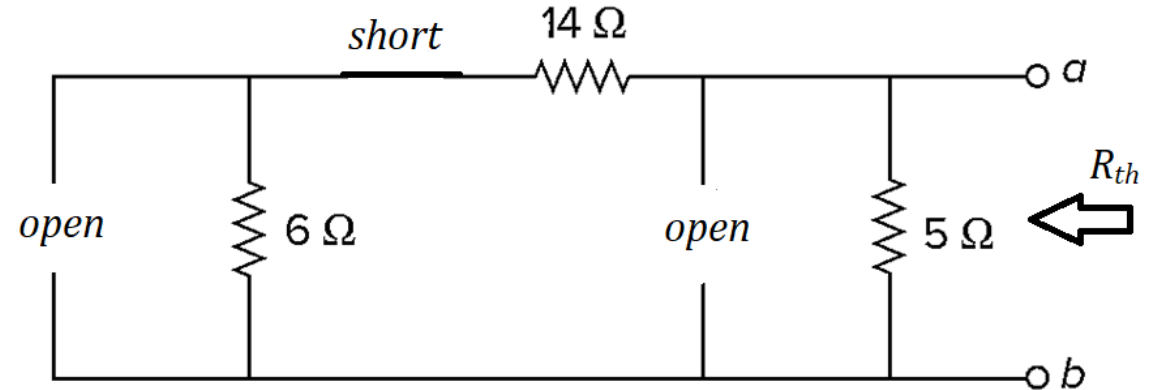
NUMERICAL

Soln:

Step 1: Finding $R_{Th/N}$. Replace current sources by open-circuits and voltage source by short-circuit.

$$R_{Th} = 5 \parallel (14 + 6) = \frac{20 \times 5}{20 + 5}$$

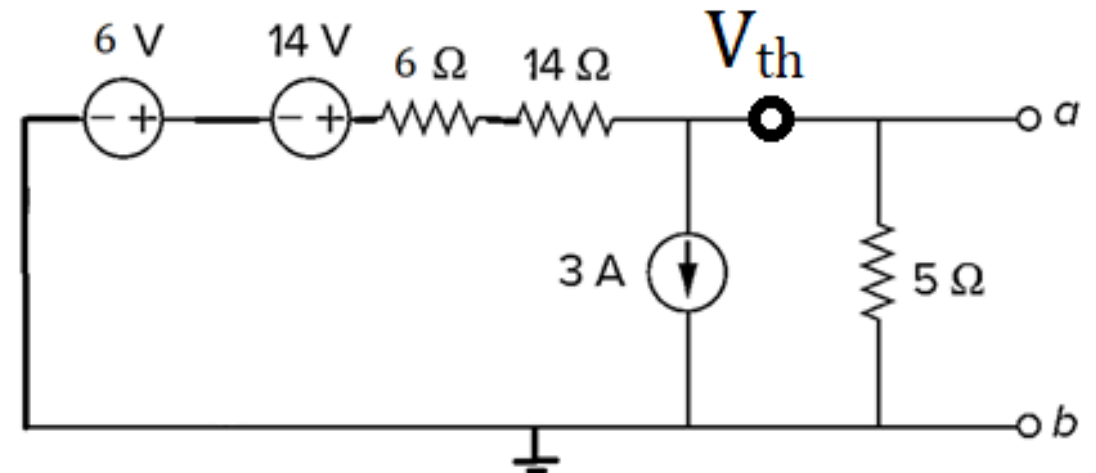
$$\Rightarrow R_{Th} = 4 \Omega = R_N$$



Step 2: Finding V_{Th} . Apply source transformation to 1A current source and apply nodal analysis.

$$\frac{14 + 6 - V_{Th}}{14 + 6} = 3 + \frac{V_{Th}}{5}$$

$$\Rightarrow V_{Th} = -8V$$



NUMERICAL

And,

$$I_N = \frac{V_{Th}}{R_{Th}} = \frac{-8}{4}$$

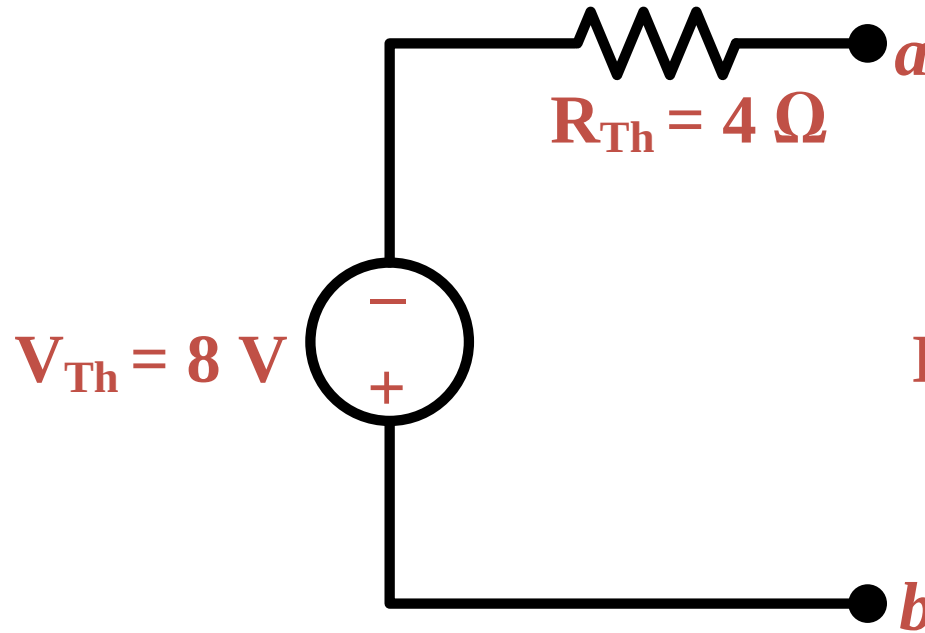
$$\Rightarrow I_N = -2A$$

Therefore,

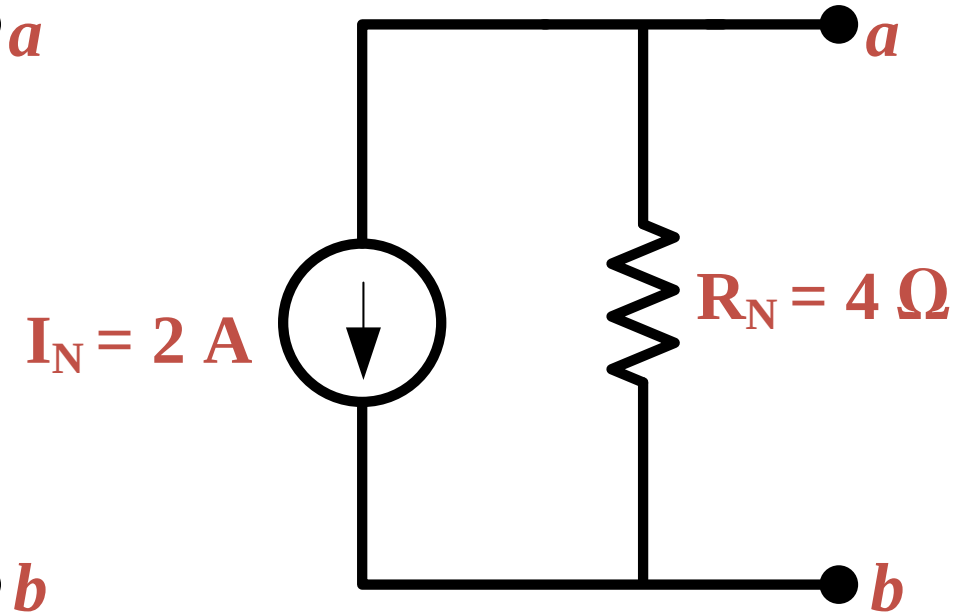
$$R_{Th} = R_N = 4\Omega$$

$$V_{Th} = -8V$$

$$I_N = -2A$$



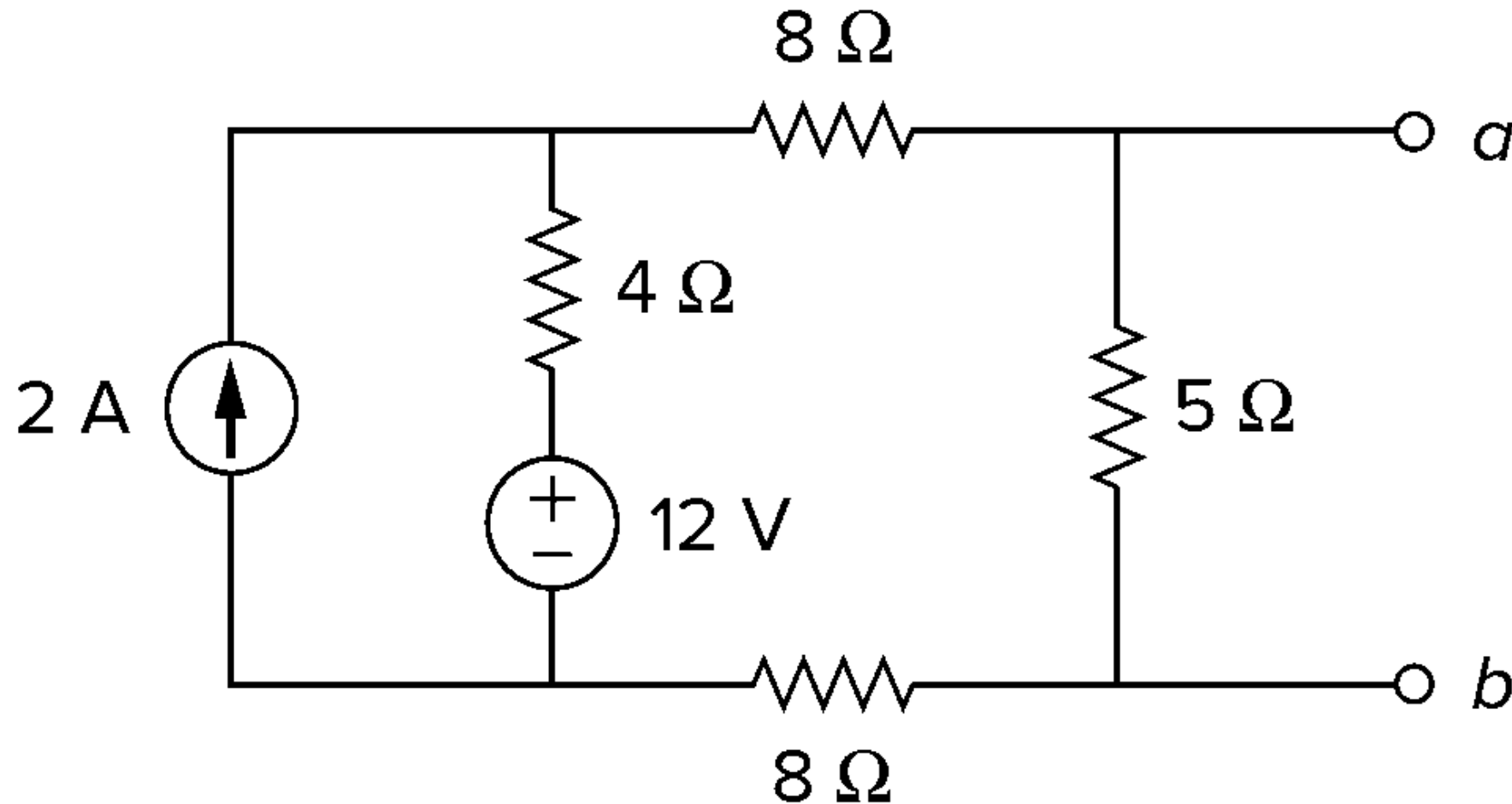
Thevenin
Equivalent Circuit



Norton Equivalent
Circuit

NUMERICAL

Q6. Find the Norton equivalent at terminals a - b of the circuit shown.



NUMERICAL

Soln:

Step 1: Finding R_N .

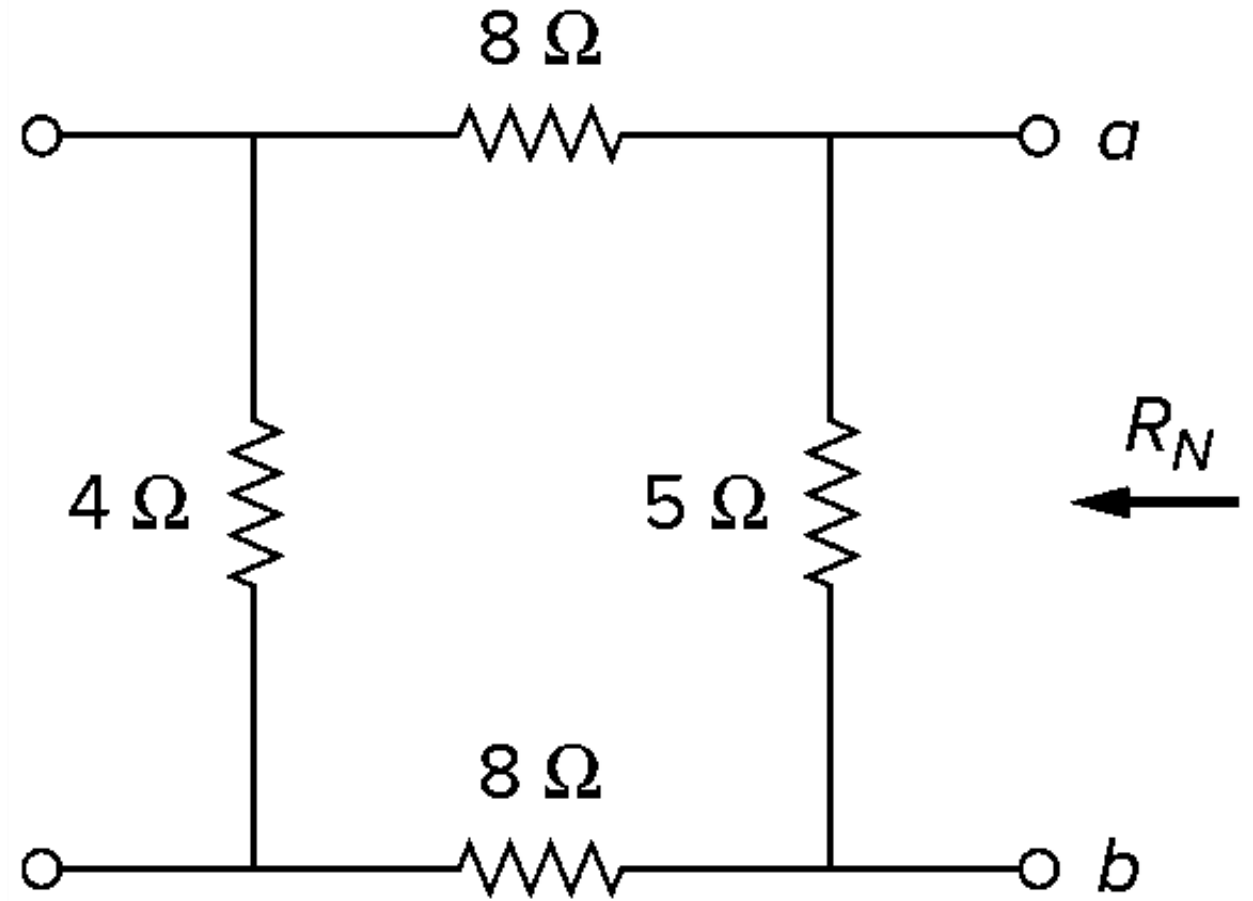
Set the independent sources to zero. This leads to the reduced circuit shown.

Thus,

$$R_N = 5 \parallel (8 + 4 + 8)$$

$$\Rightarrow R_N = 5 \parallel 20 = \frac{20 \times 5}{20 + 5}$$

$$\Rightarrow \boxed{R_N = 4\Omega}$$



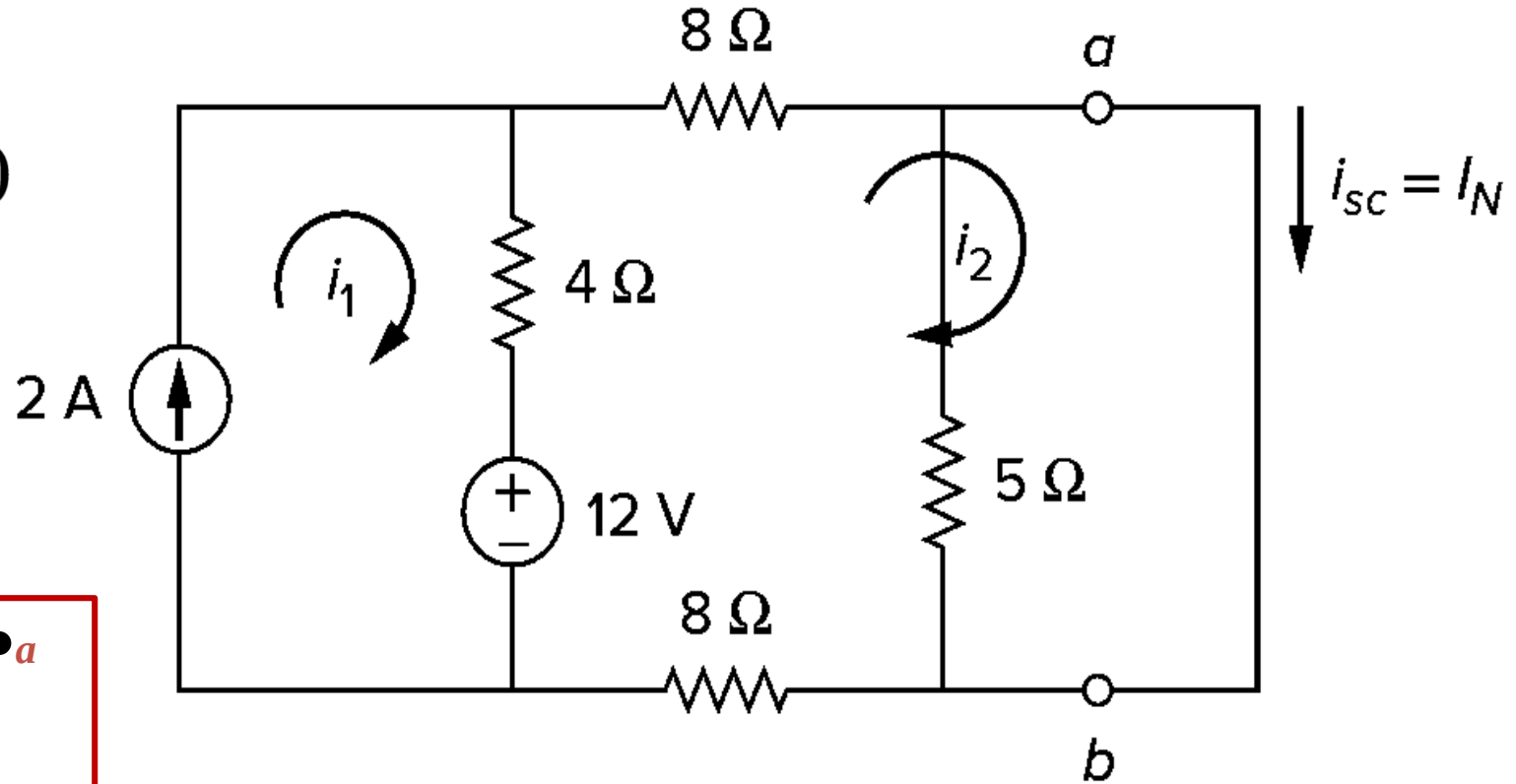
NUMERICAL

Step 2: Finding I_N . Short-circuit terminals a and b .
Ignore the $5\text{-}\Omega$ resistor because it has been short-circuited.

Applying mesh analysis,
 $i_1 = 2\text{A}$; $20i_2 - 4i_1 - 12 = 0$

On solving, we get

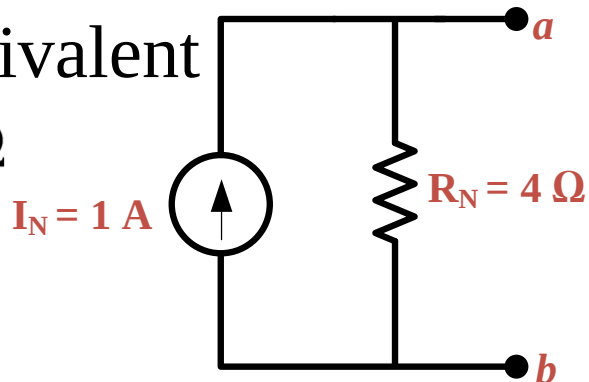
$$i_2 = \boxed{1\text{A}} = i_{sc} = \boxed{I_N}$$



Norton's Equivalent

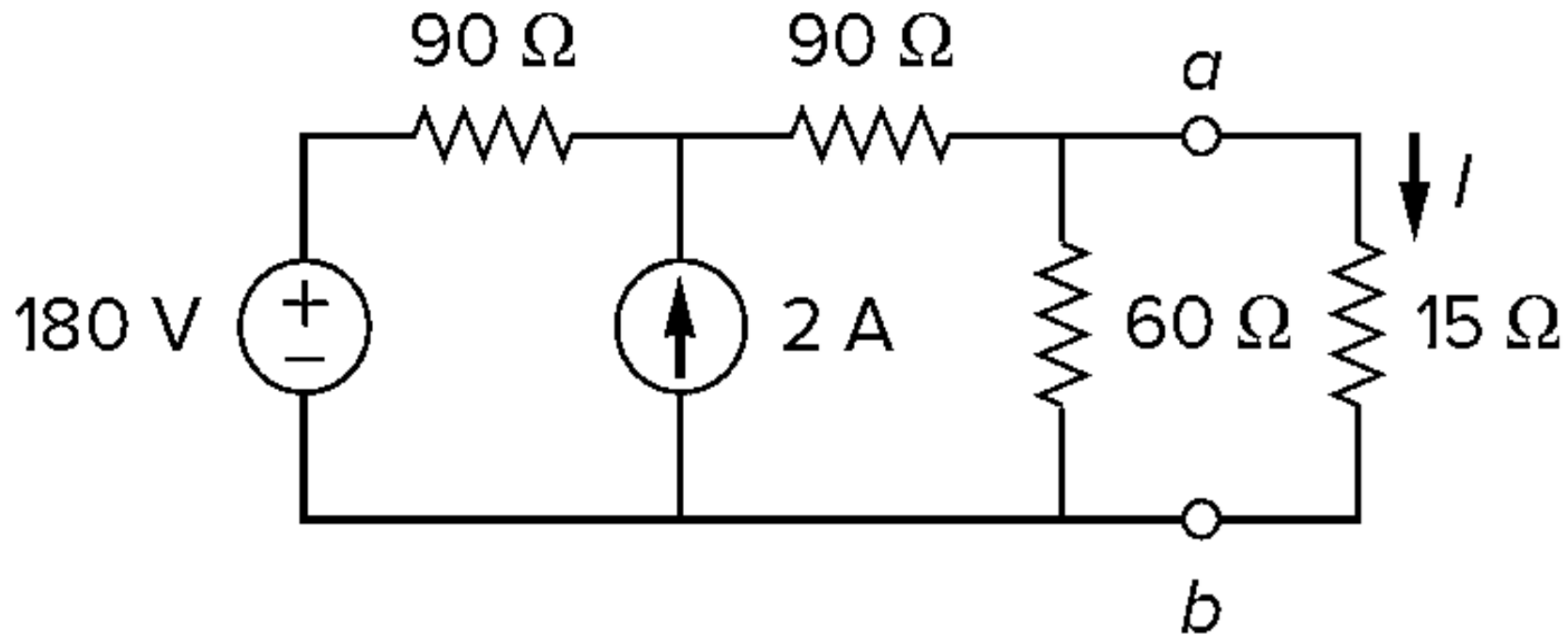
$$R_{Th} = R_N = 4\Omega$$

$$I_N = 1\text{A}$$



UNSOLVED NUMERICAL

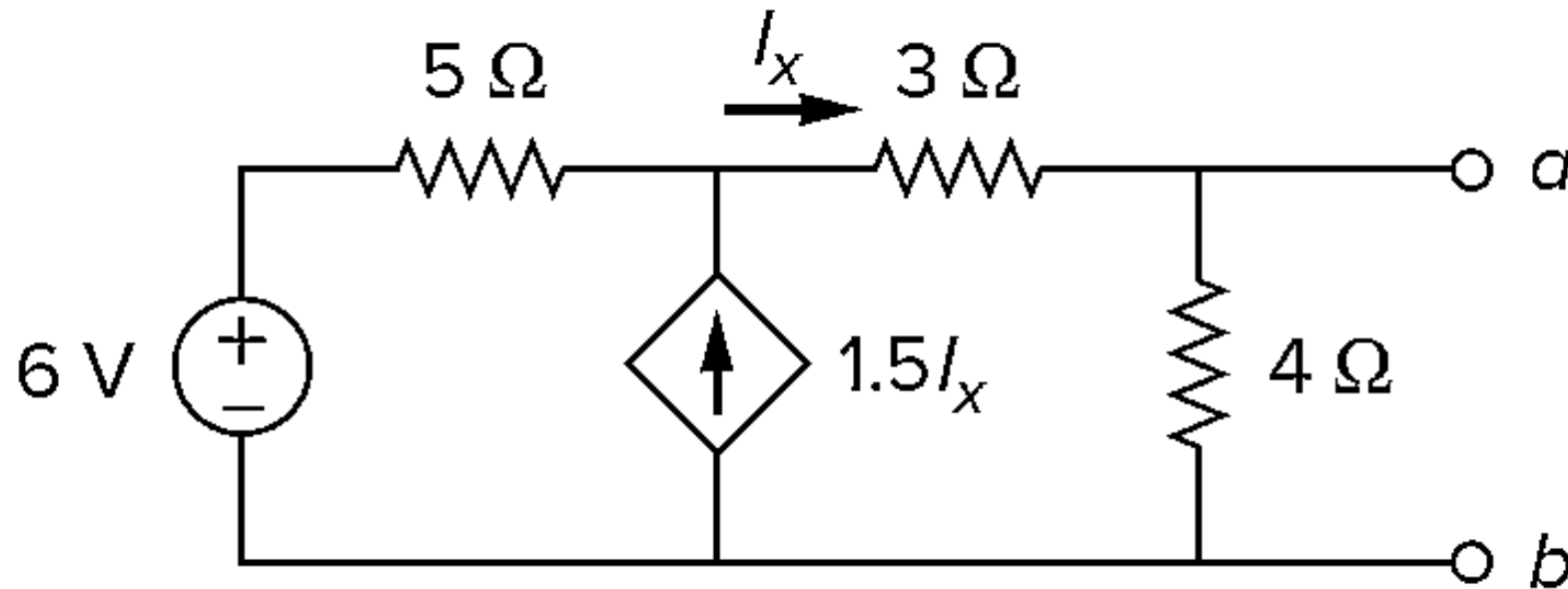
Q1. Using Thevenin's theorem, find the equivalent circuit to the left of the terminals in the circuit below. Then find I .



Ans: $V_{Th} = 90\text{ V}$, $R_{Th} = 45\ \Omega$, $I = 1.5\text{ A}$.

UNSOLVED NUMERICAL

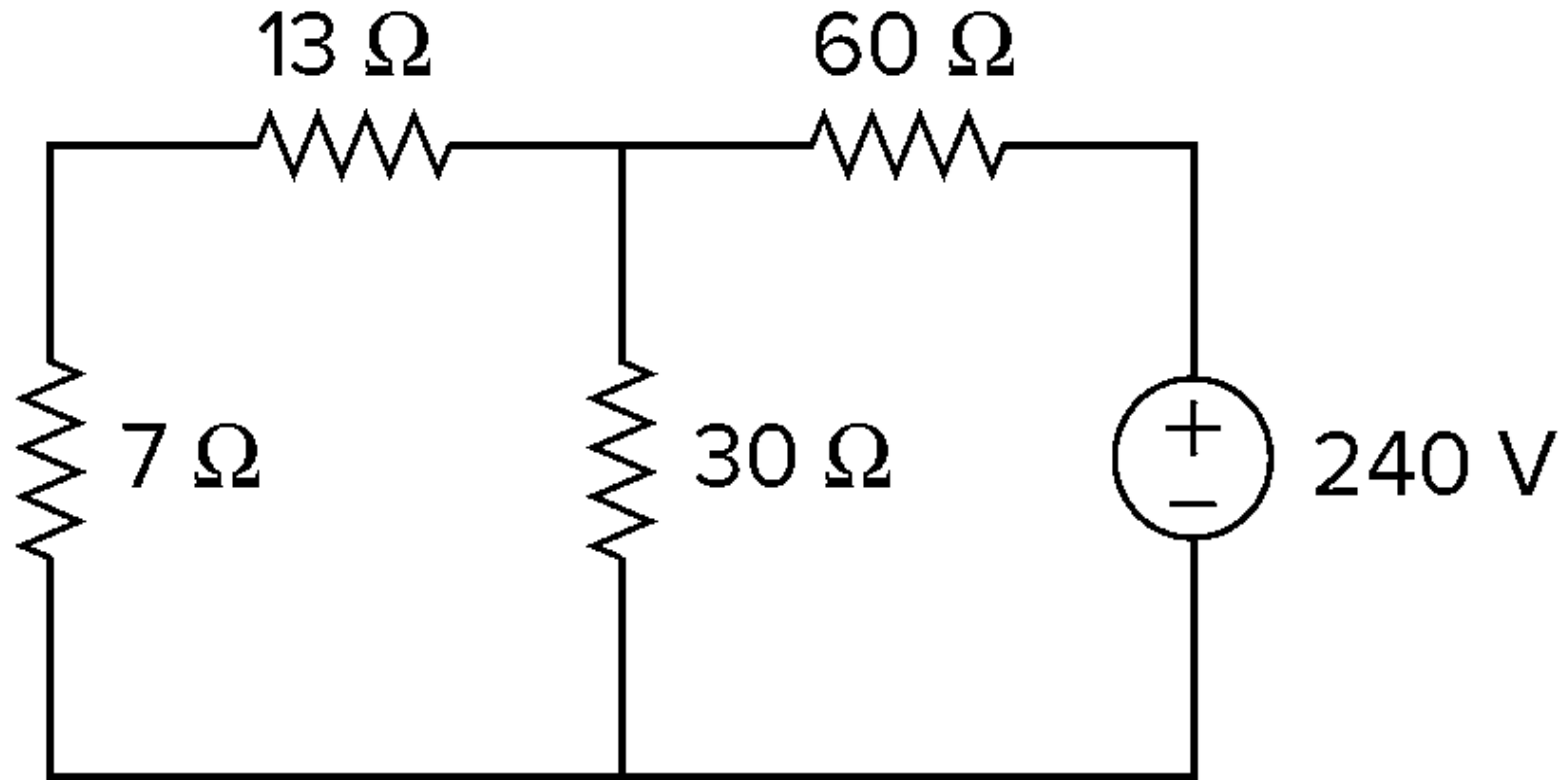
Q2. Find the Thevenin equivalent circuit of the circuit shown below to the left of the terminals.



Ans: $V_{Th} = 5.333 \text{ V}$, $R_{Th} = 444.4 \text{ m}\Omega$.

UNSOLVED NUMERICAL

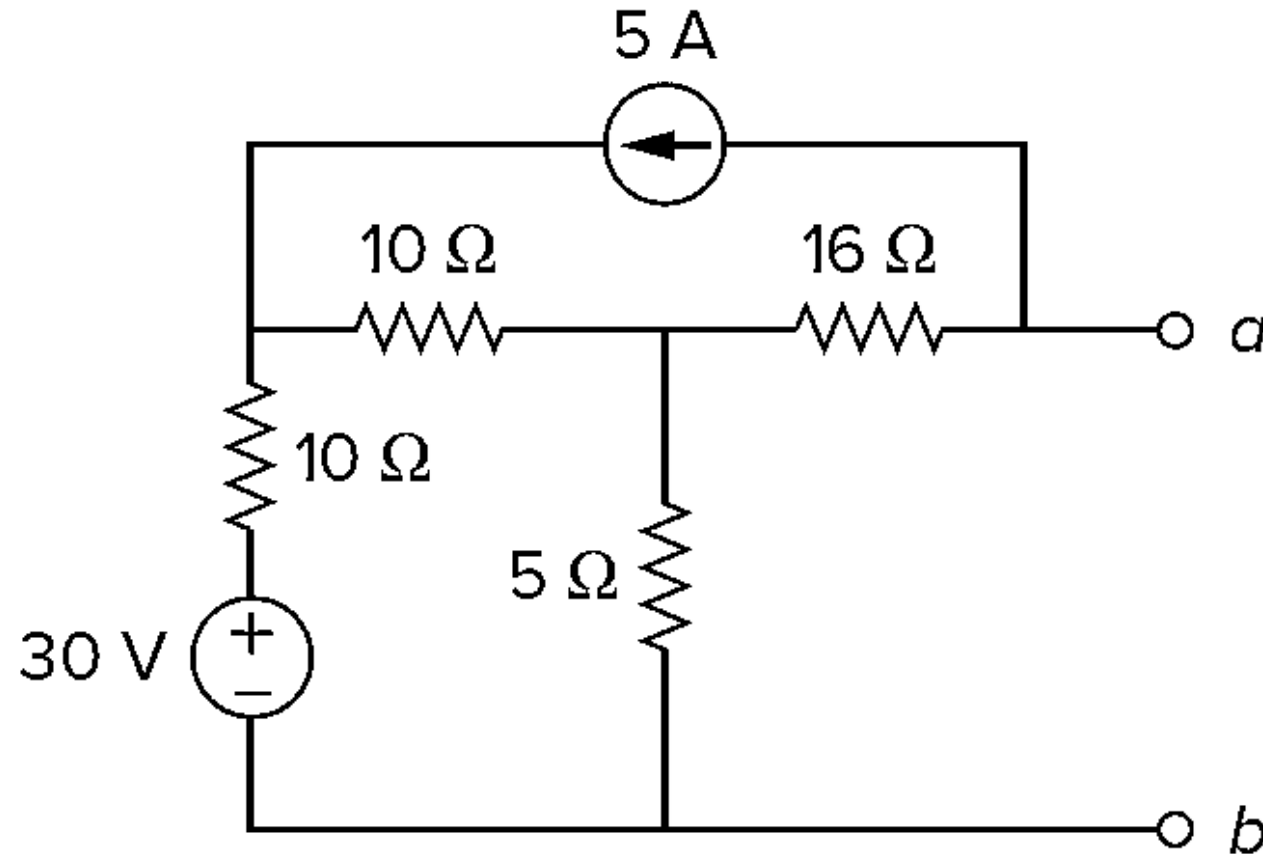
Q3. Determine the Thevenin equivalent circuit shown below, as seen by the 7Ω resistor. Then calculate the current flowing through the 7Ω resistor.



Ans: $V_{Th} = 80\text{ V}$, $R_{Th} = 9.069\Omega$.

UNSOLVED NUMERICAL

Q4. Obtain the Thevenin equivalent at terminals a-b of the circuit shown below.



Ans: $V_{Th} = 38 \text{ V}$, $R_{Th} = 20 \text{ } \Omega$.